



*Better lives through livestock*

# Pathogen Genomic Identification & Surveillance (Rift Valley fever Virus)

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*Project: **ILRI Genomics Platform***

*Program: **Health***

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*Online documentation: [github/kibet-gilbert/rvf-v-ngs-surveillance](https://github.com/kibet-gilbert/rvf-v-ngs-surveillance)*

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# Overview

- Genomic surveillance & Why it matters for emerging pathogens
- Recovering viral genomes from clinical samples 
- NGS approaches (Metagenomics vs. Targeted Enrichment)
- Targeted Enrichment through Tiled amplicon sequencing
-  Real-world application: RVFV, Dengue, and Chikungunya
- Lineage assignment and phylogenetic analysis
-  Impact on outbreak response and surveillance

# Pathogen Identification and Surveillance Through Genomics

# Why Genomic Surveillance?

- Systemic sequencing
- Over time & Place
- Linkage to epidemiological data



**Accurate pathogen identification**  
as compared to  
convention  
diagnostic tools.



**Track transmission  
and evolutionary  
changes**  
over time and  
geography.



**Predict the  
potential severity**  
of new pathogens  
or variants.



**Valuable  
information for  
drug development**  
for vaccines and  
therapeutics.

# Rift Valley fever Virus

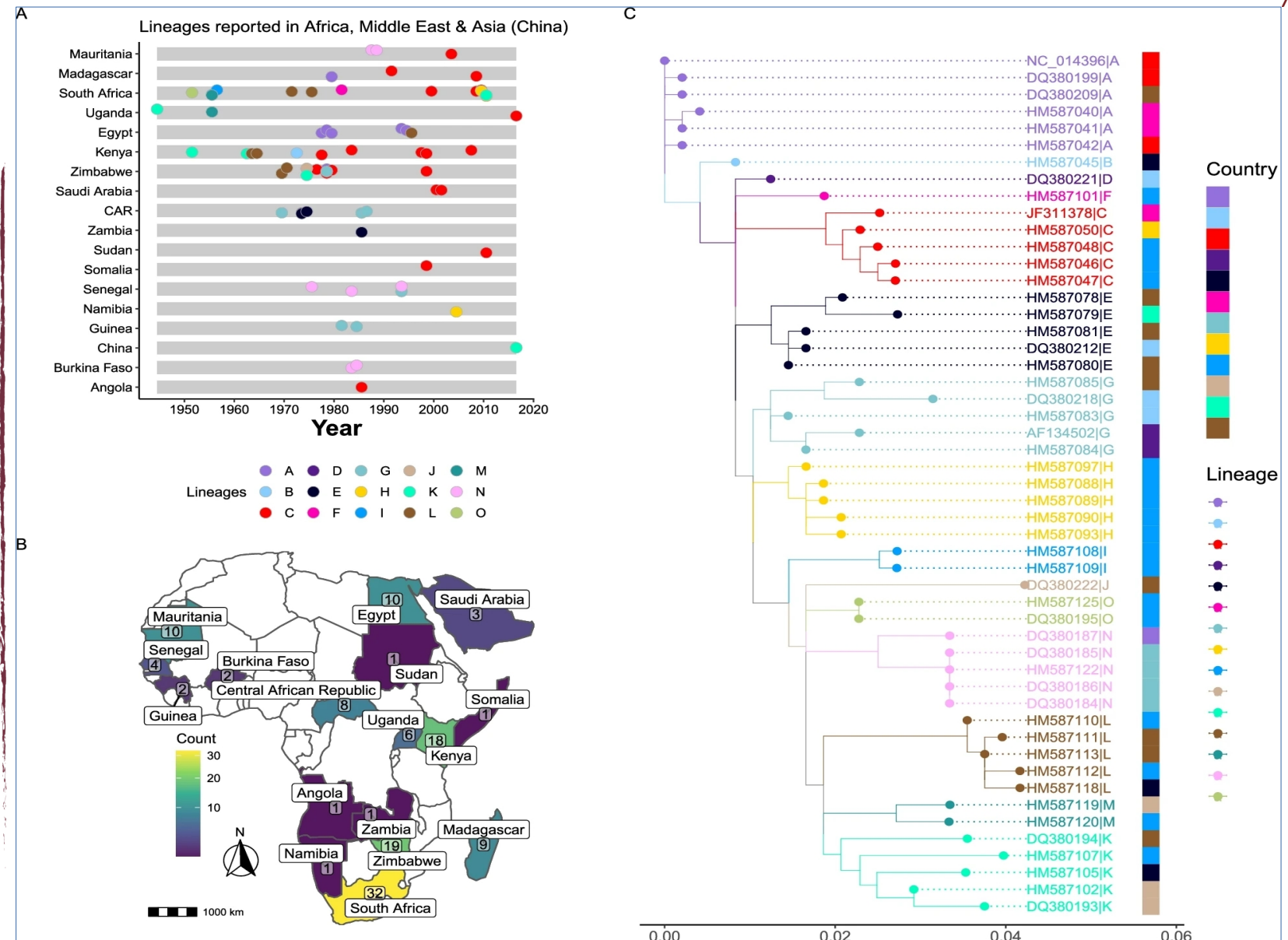
## Rift Valley fever Virus (*Phlebovirus riftense*)

- First described 1931 - sheep abortion epidemic in Kenya
- Endemic in sub-Saharan Africa & Arabian Peninsula
- Aedes & Culex mosquitoes
- Documented Epi-zootic & Epidemic events: Egypt (1977), Western Africa (1988) and Arabian Peninsula (2000)
- Humans: 0.5-2.0% fatality ratio
- Livestock: "Abortion Storm" - neonatal mortality (50-100%)



# RVFV Distribution

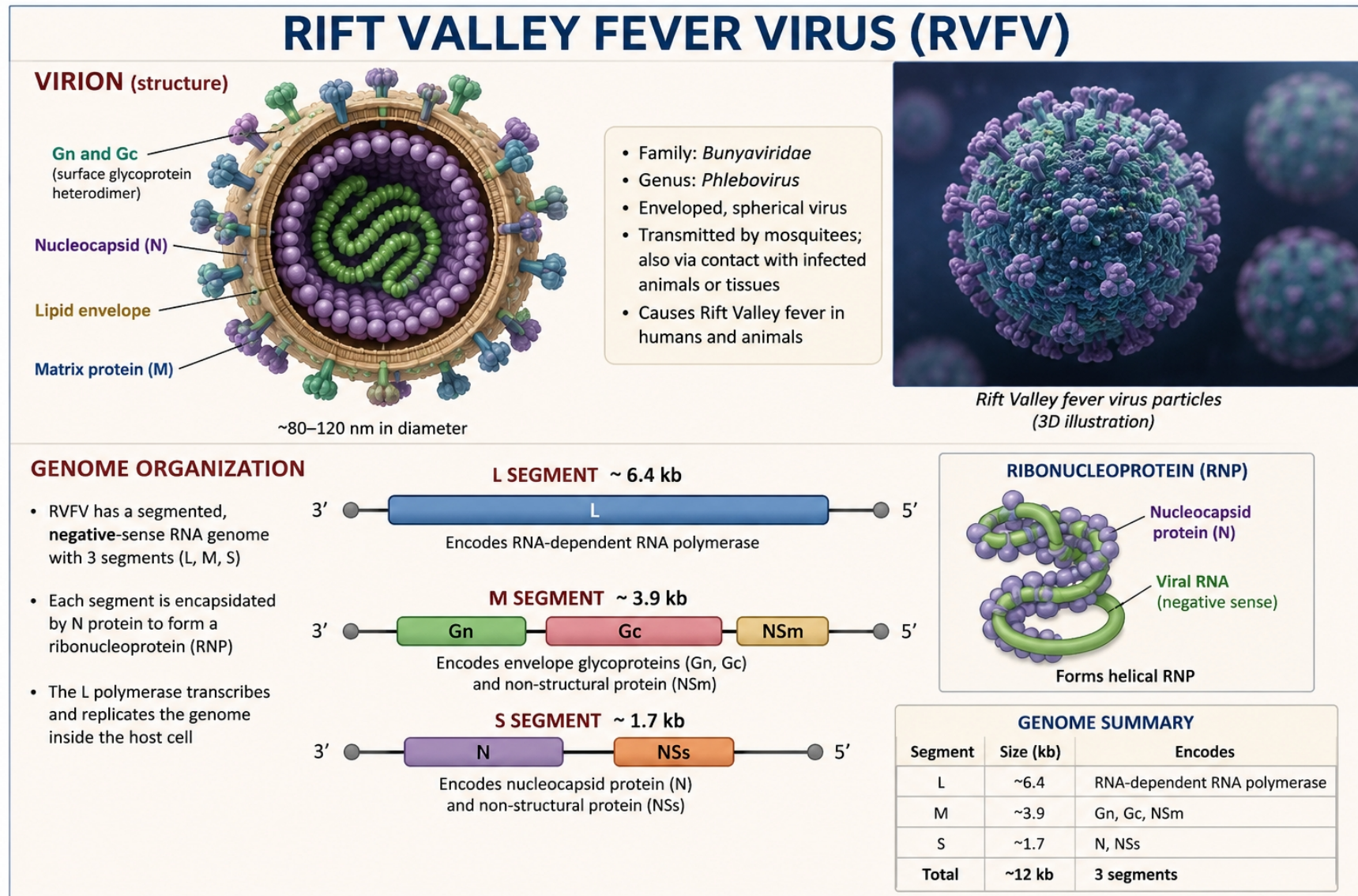
- Lineages: 15 distinct lineages identified in Africa (A through O)
- Lineage C: Most predominant (documented since 1976)
- Lineage H: Second most common in recent outbreaks
- Other lineages: Geographically or temporally restricted



Juma, J., Fonseca, V., Konongoi, S.L. et al. Genomic surveillance of Rift Valley fever virus: from sequencing to lineage assignment. BMC Genomics 23, 520 (2022).

# RVFV Genome Organization

- Negative-sense, ssRNA segments
- Low genetic diversity overall.
- 4% nucleotide-level variation (highest: ~5% in M-segment)
- 1% amino acid-level variation
- NO well-defined serotypes

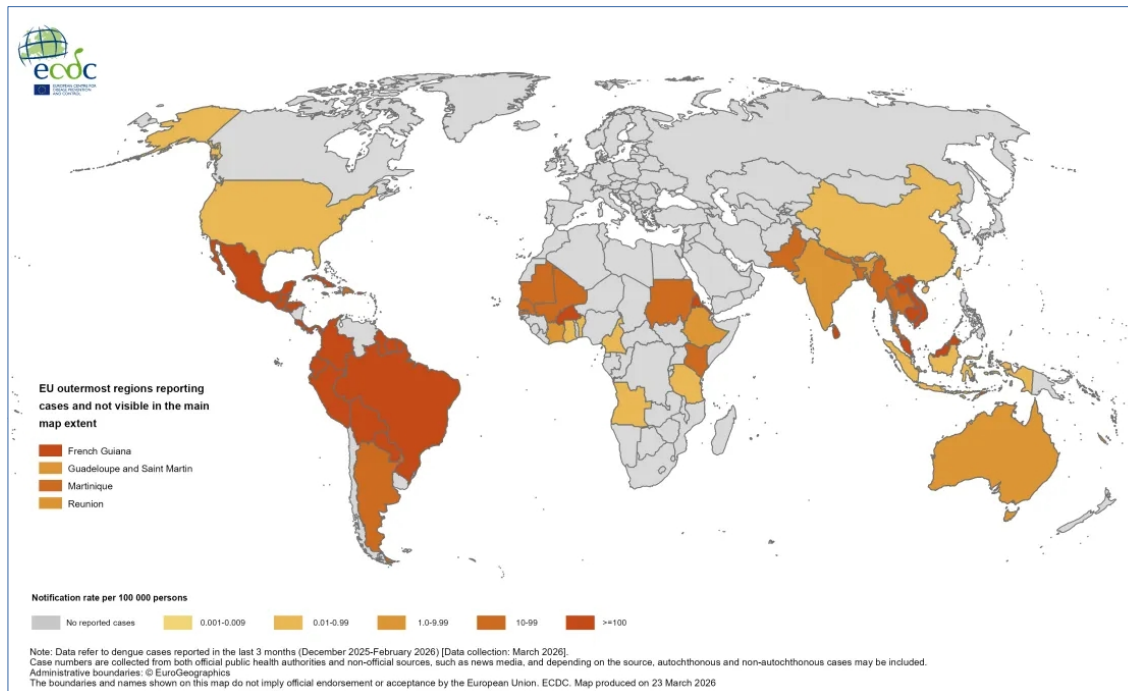


| GENOME SUMMARY |               |                              |
|----------------|---------------|------------------------------|
| Segment        | Size (kb)     | Encodes                      |
| L              | ~6.4          | RNA-dependent RNA polymerase |
| M              | ~3.9          | Gn, Gc, NSm                  |
| S              | ~1.7          | N, NSs                       |
| <b>Total</b>   | <b>~12 kb</b> | <b>3 segments</b>            |

# Compared to DENV and CHIKV

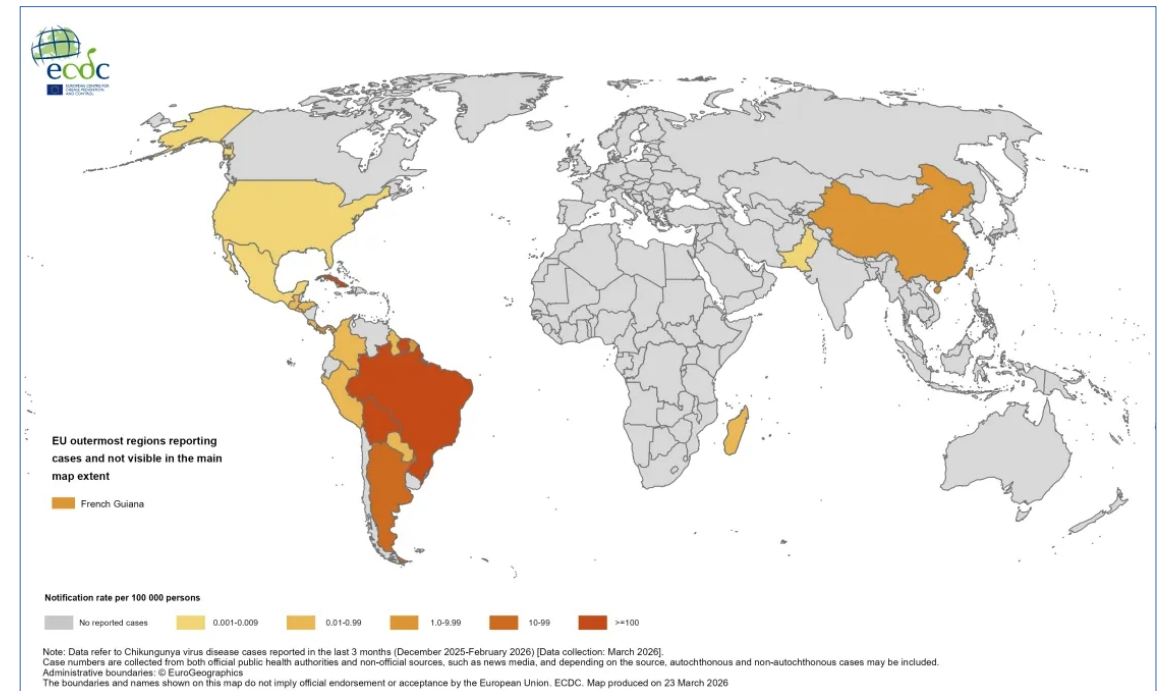
## Dengue Virus

- 4 serotypes (DENV-1 to DENV-4)
- +Single-stranded RNA genome (~10.7-11 kb)
- WHO: Top 10 global public health threats



## Chikungunya Virus

- • Genome: ~11.8 kb +single-stranded RNA
- • Characteristic debilitating joint pain (arthralgia)
- • Endemic in >110 countries; significant African circulation

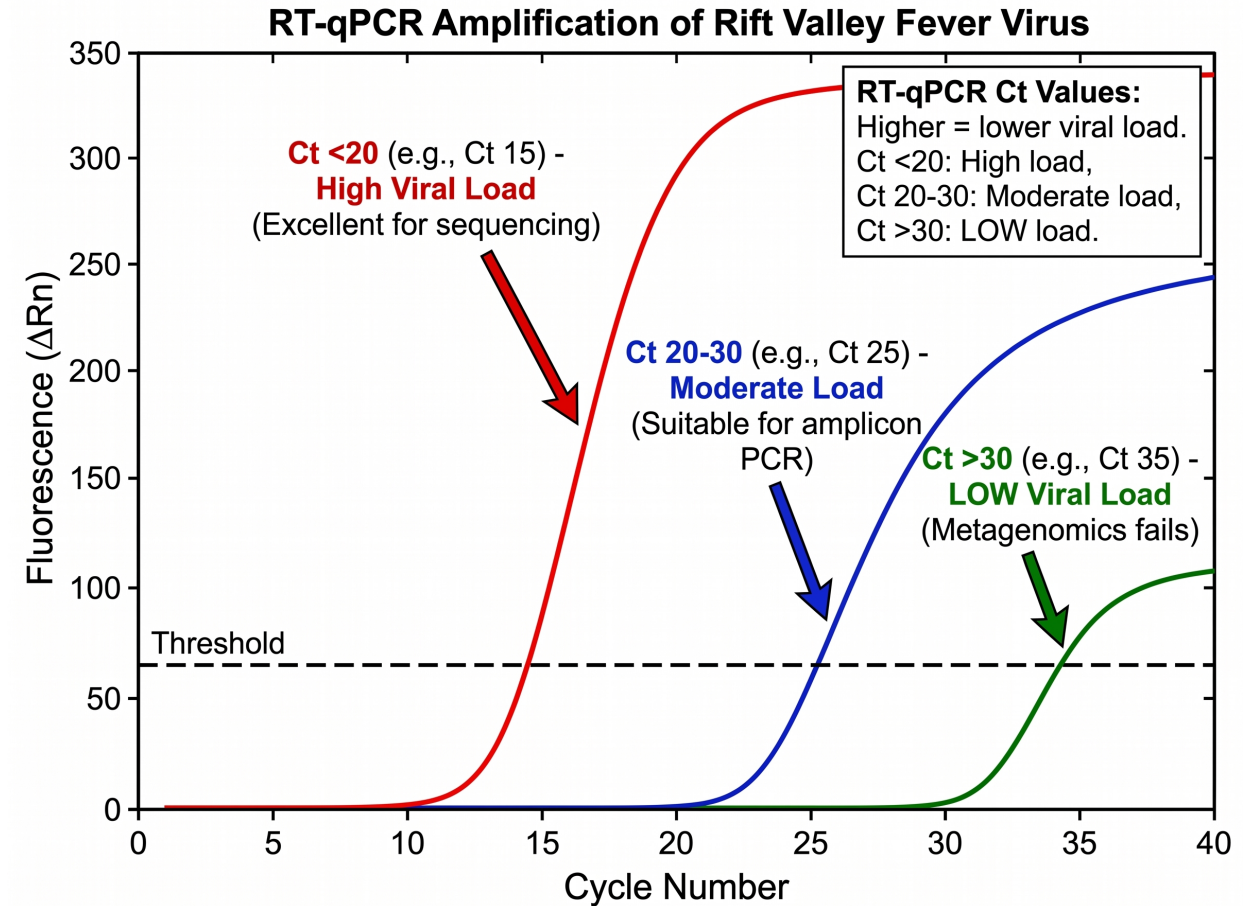


# NGS Approaches

# NGS Challenges: Low Viral Titres

RT-qPCR Ct values: Higher = lower viral load

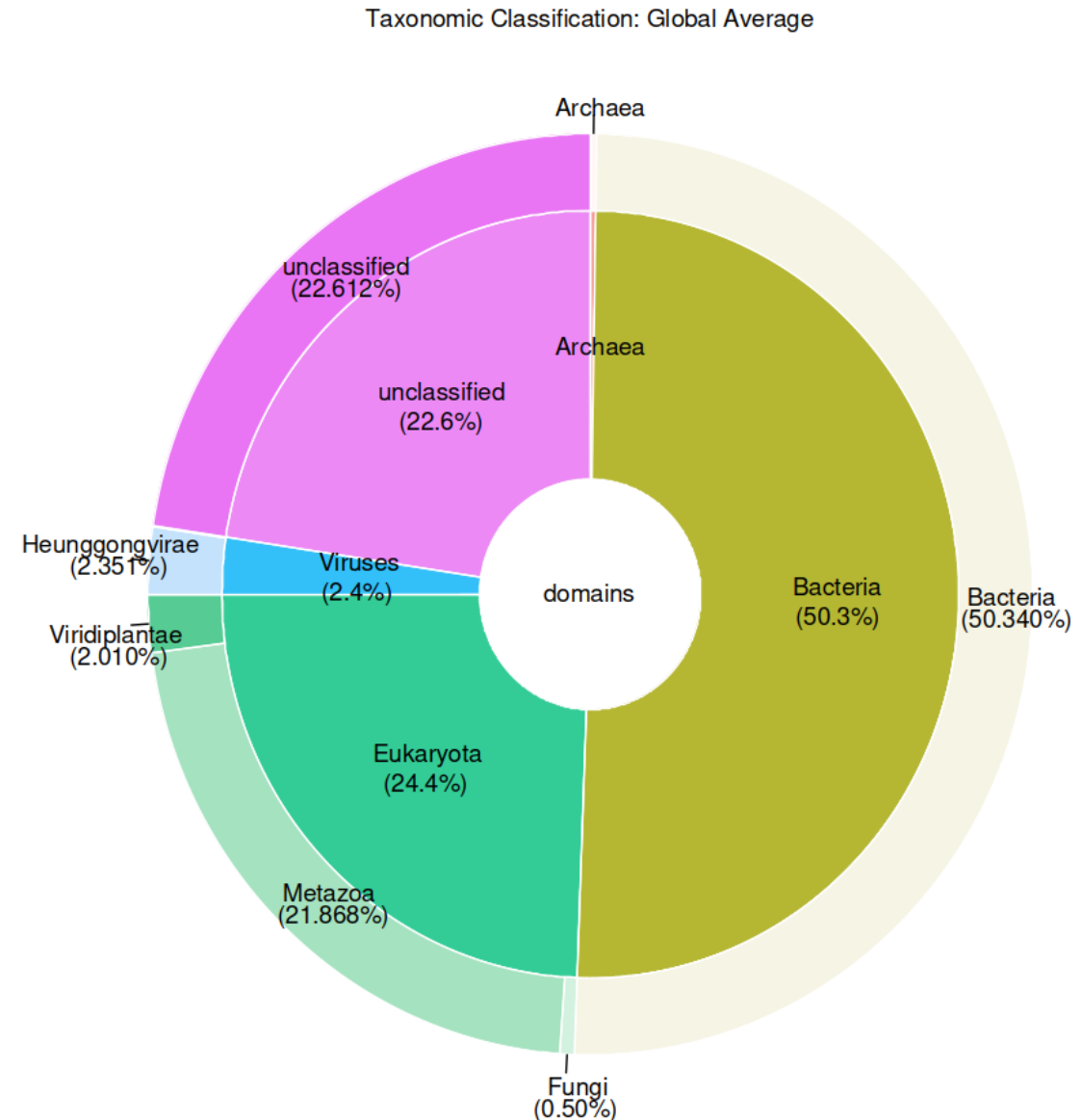
- Ct <20: High viral load (excellent for sequencing)
- Ct 20-30: Moderate load (suitable for amplicon PCR)
- Ct >30: LOW viral load (metagenomics fails)



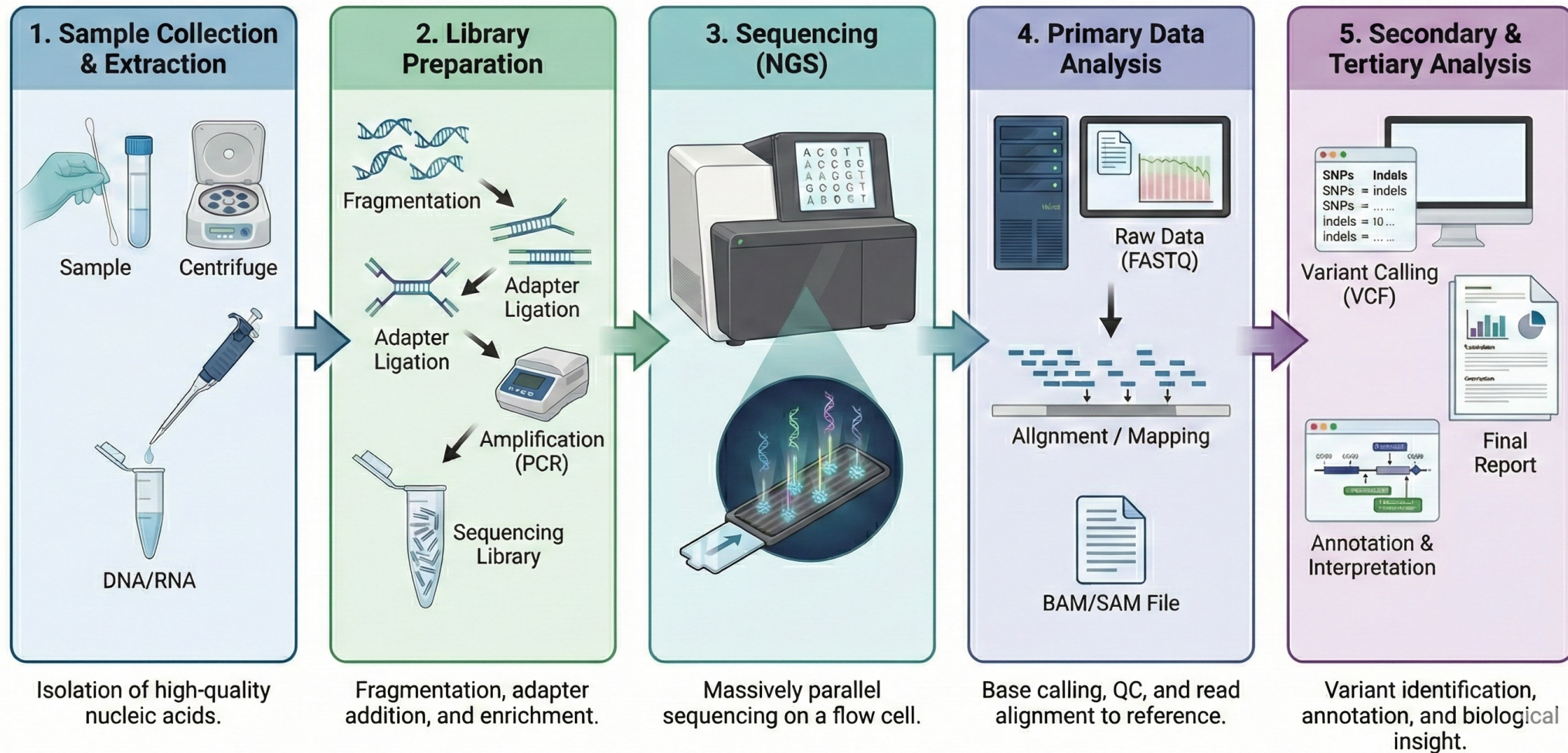
# NGS Challenges: RVFV

RVFV Peak viremia in first 5-8 days post-infection

- After 8 days, **viremia drops** dramatically
- Short window for **sequence recovery** from clinical samples
- Host and host-microbiota **contamination**
- **Sample Degradation:** Viral RNA is highly unstable and can degrade quickly if not handled correctly



# NGS Overview



# NGS Approach 1: Metagenomics

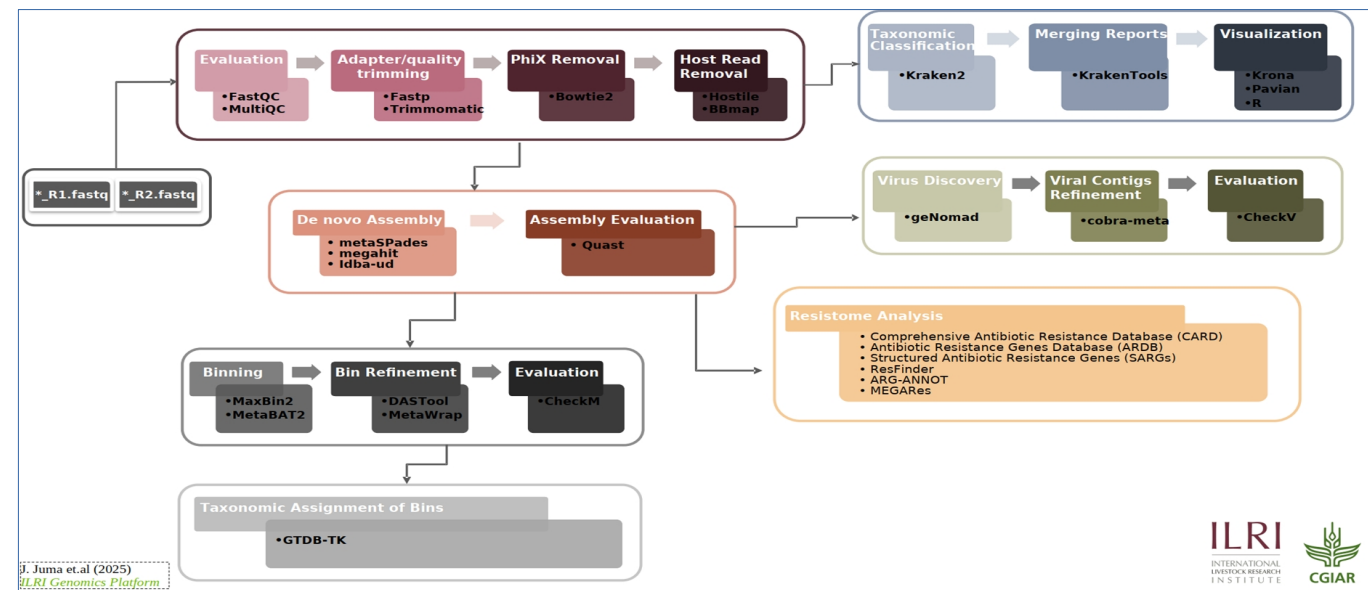
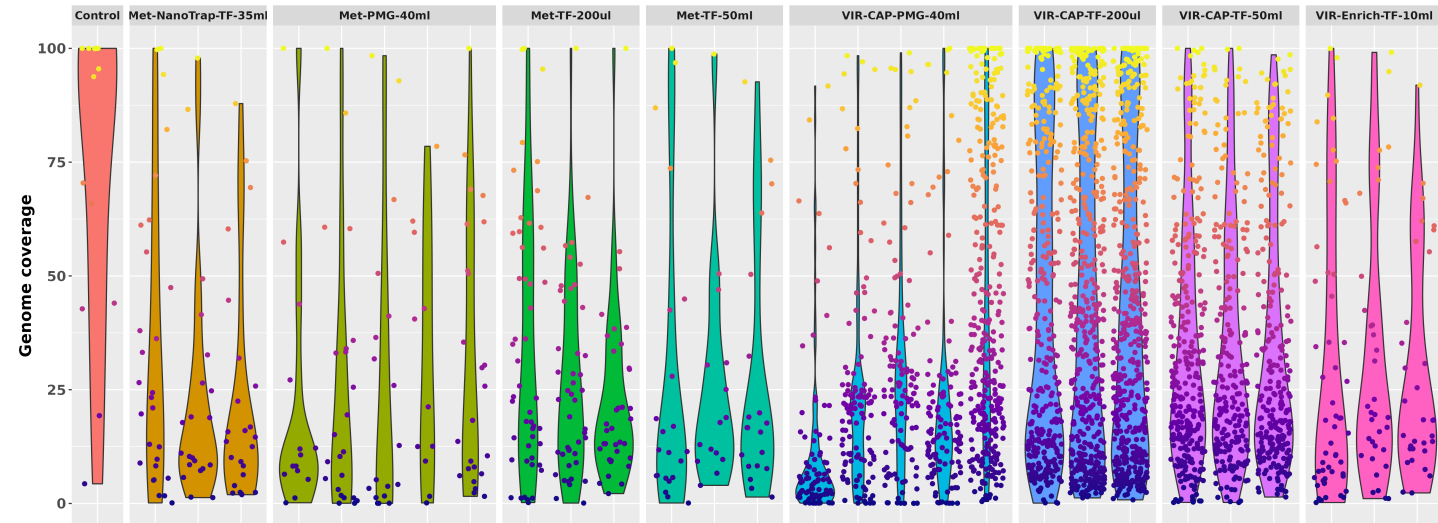
**Concept:** Sequence Everything in a Sample

## Advantages:

- Unbiased - discovers unexpected pathogens
- Useful when etiology UNKNOWN

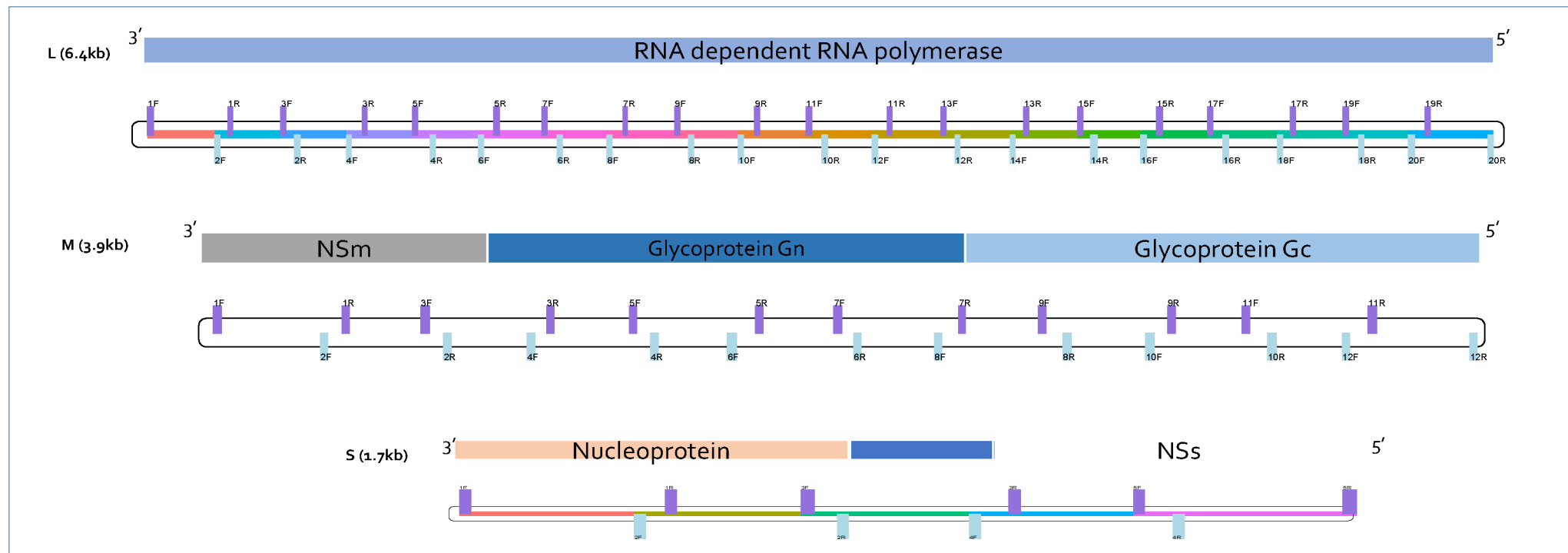
## Disadvantages:

- Low sensitivity for low-abundance pathogens
- Expensive (many reads wasted on host)
- Slow analysis (extensive bioinformatic processing)
- Requires high coverage (100M+ reads)



# NGS Approach 2: Tiled Amplicons Sequencing

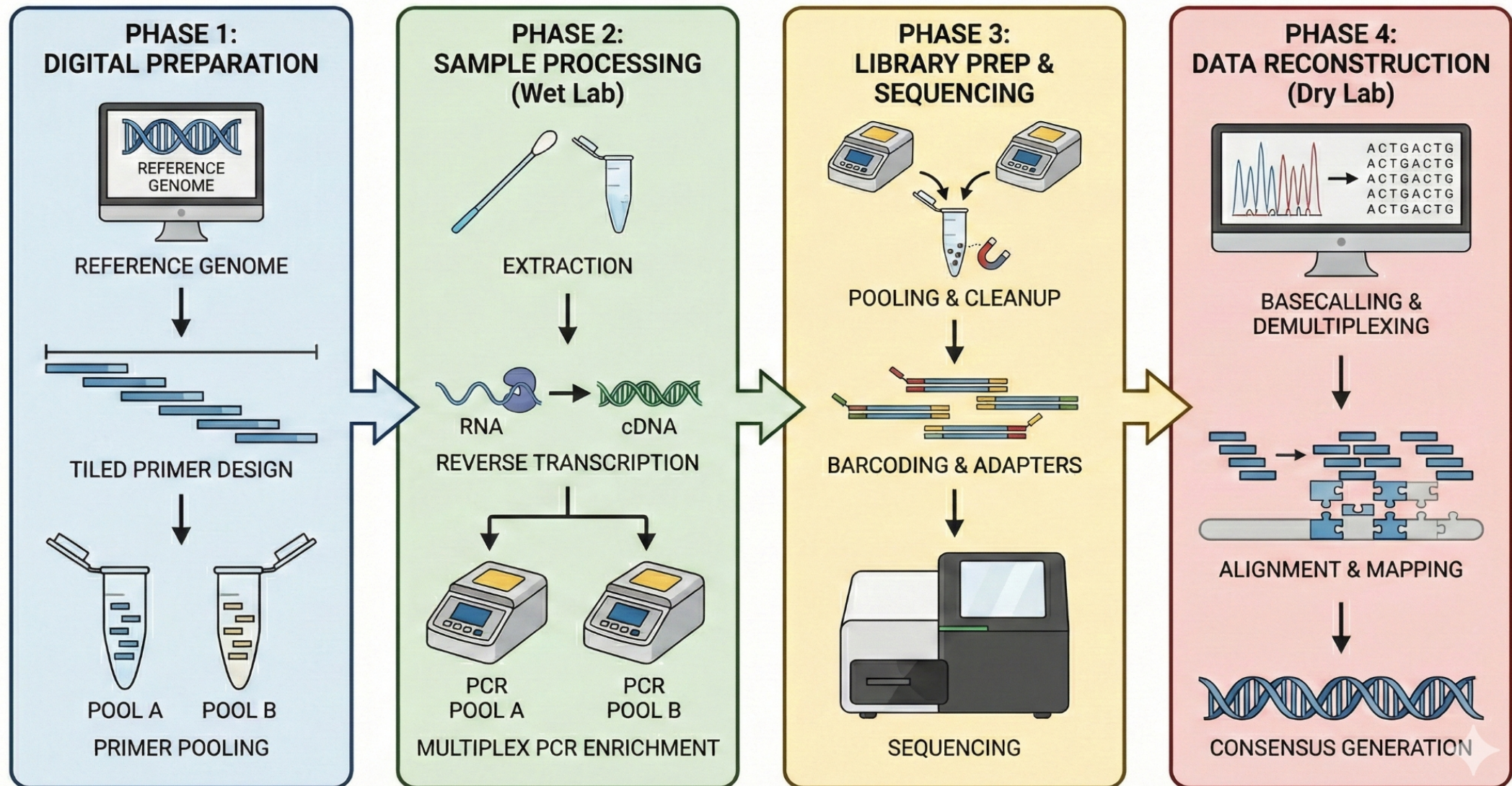
- **Target enrichment** method using multiplex PCR
- Primer pools (“Tiles”) - amplicons overlap - complete coverage of the target genome.



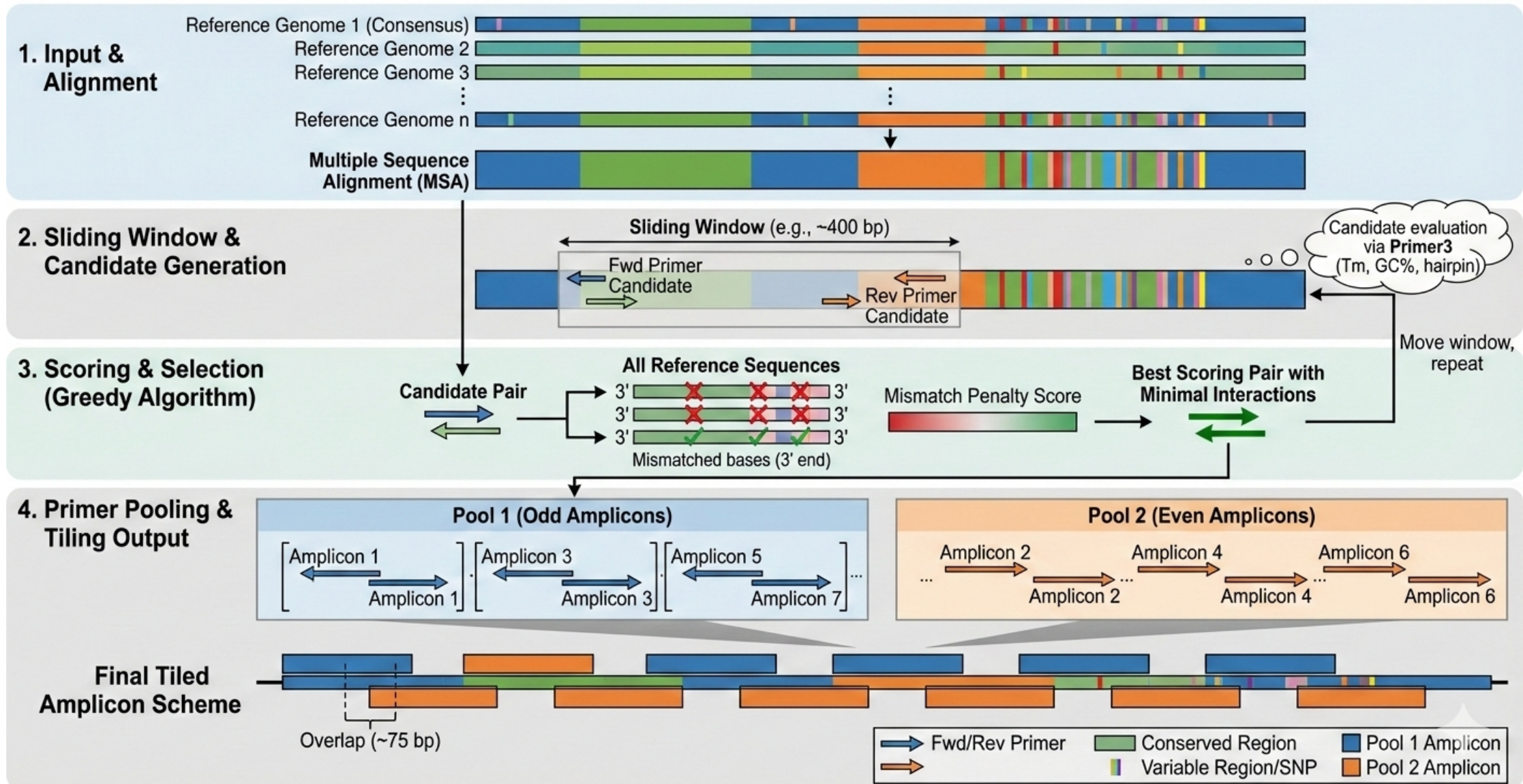
Juma, J.; Konongoi, S.L.; Nsengimana, I.; et al. Using Multiplex Amplicon PCR Technology to Efficiently and Timely Generate Rift Valley Fever Virus Sequence Data for Genomic Surveillance. *Viruses* (2023).

# Tiled Amplicon Approach

# Tiled amplicons Sequencing Workflow



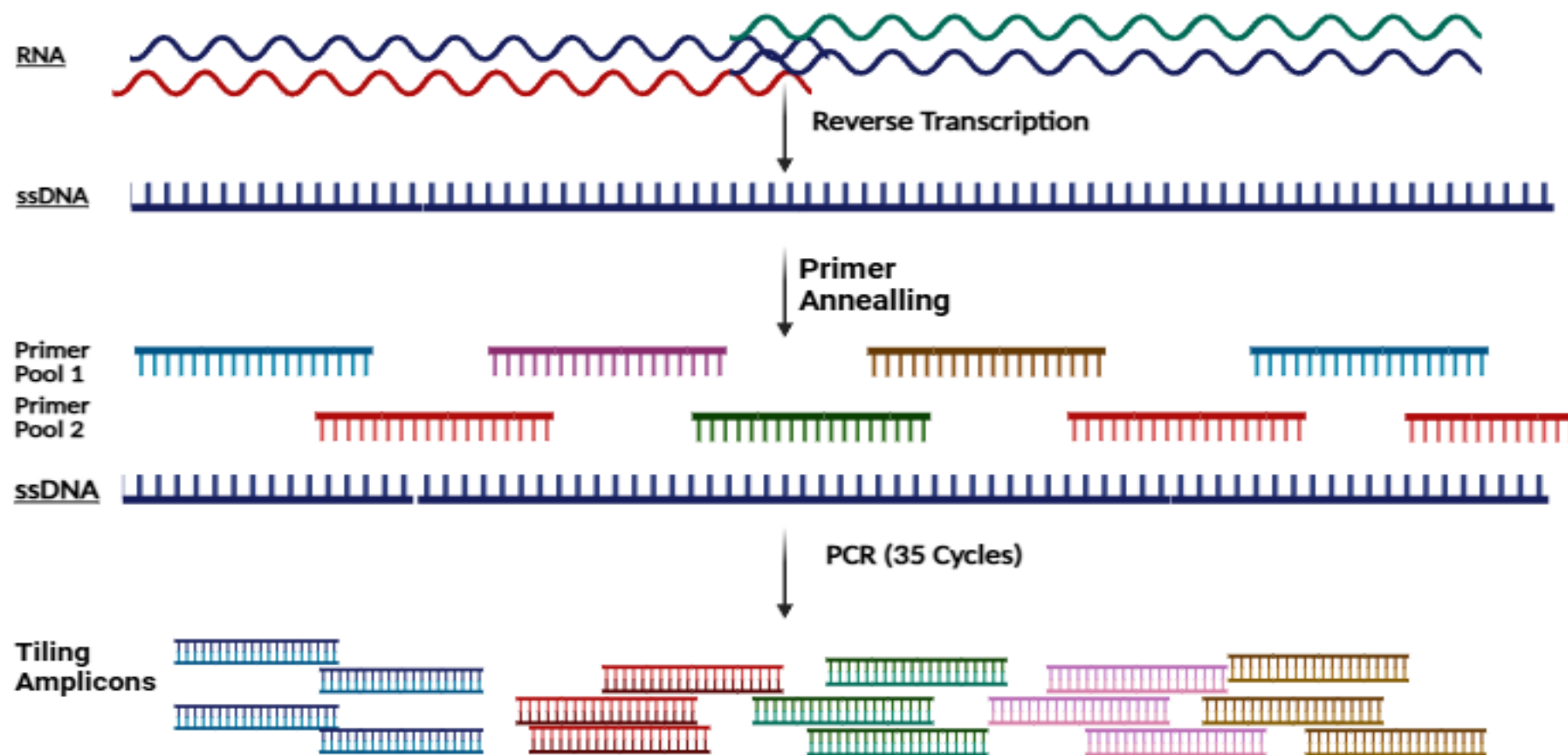
# PrimalScheme: Amplicon Tiling Primer Design from Multiple Genomes



Web App: [Primalcheme.com](http://Primalcheme.com)

Command Line [artic-network/primalscheme3](https://github.com/artic-network/primalscheme3)

# Target Enrichment Step...



# Advantages



## High Sensitivity and Low Input Requirement.

It is highly effective for samples with low viral titers.



## Direct Sequencing from Clinical Samples.

It eliminates the need for laborious laboratory viral culture.



## Superior Signal-to-Noise Ratio.

The process selectively amplifies the viral genome while ignoring background host DNA or RNA.



## Rapid Turnaround Time and Cost-effectiveness.

The protocol can be completed in 1–2 days and reduces the sequencing cost.



## Comprehensive Coverage.

The primer design ensures that the entire coding sequence is recovered for downstream analysis.

# Quality Checks & Screening

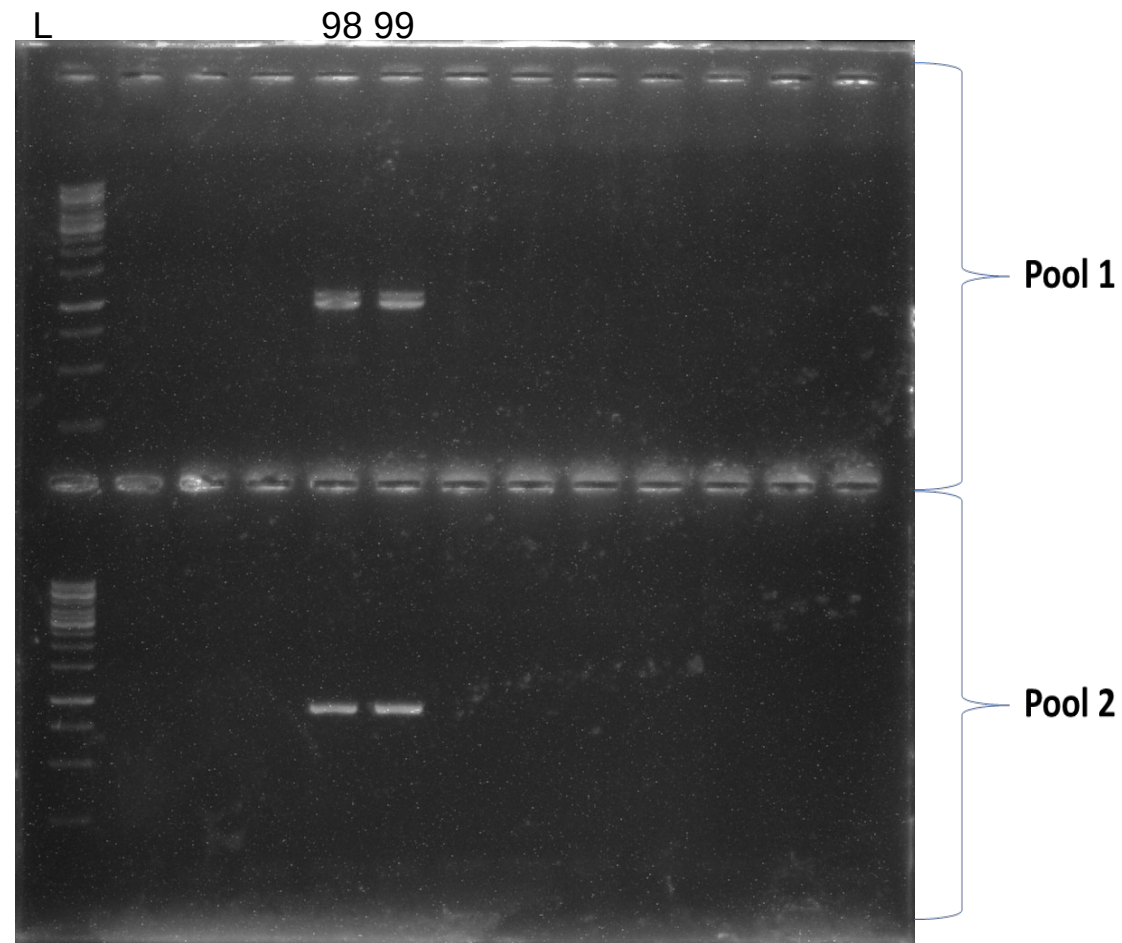
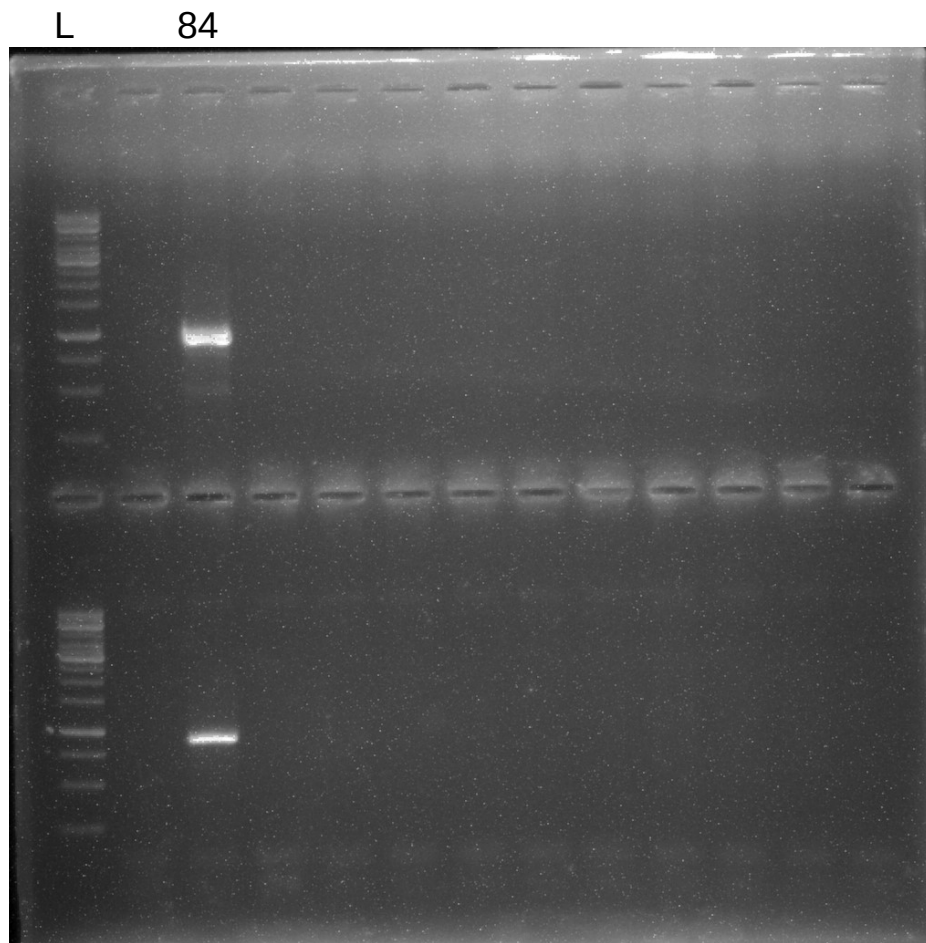
## DenV-2 Virus tiled Amplicons

GeneRuler 1 kb DNA Ladder

| bp    | ng/0.5µg | %  |
|-------|----------|----|
| 10000 | 30       | 6  |
| 8000  | 30       | 6  |
| 6000  | 70       | 14 |
| 5000  | 30       | 6  |
| 4000  | 30       | 6  |
| 3500  | 30       | 6  |
| 3000  | 70       | 14 |
| 2500  | 25       | 5  |
| 2000  | 25       | 5  |
| 1500  | 25       | 5  |
| 1000  | 60       | 12 |
| 750   | 25       | 5  |
| 500   | 25       | 5  |
| 250   | 25       | 5  |

1% agarose

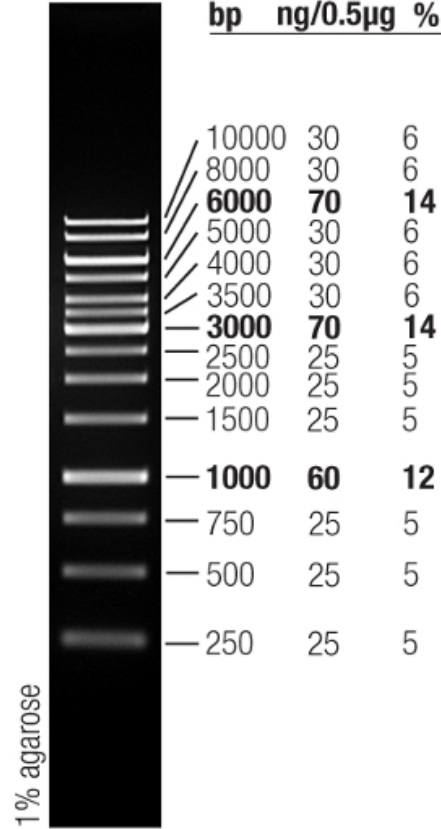
0.5 µg/lane, 8 cm length gel,  
1X TAE, 7 V/cm, 45 min



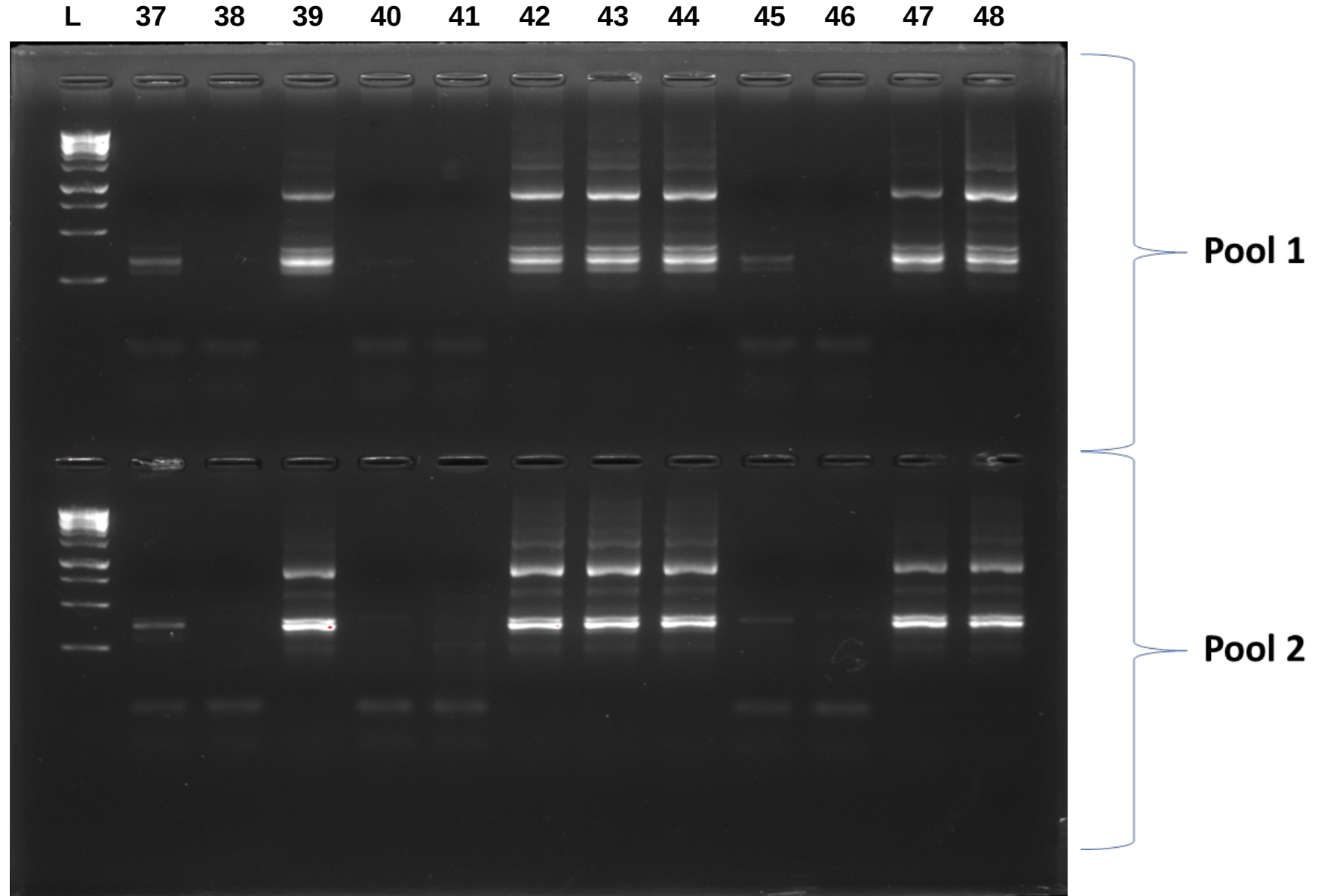
# Chikungunya Virus tiled Amplicons

GeneRuler 1 kb DNA Ladder

bp ng/0.5µg %



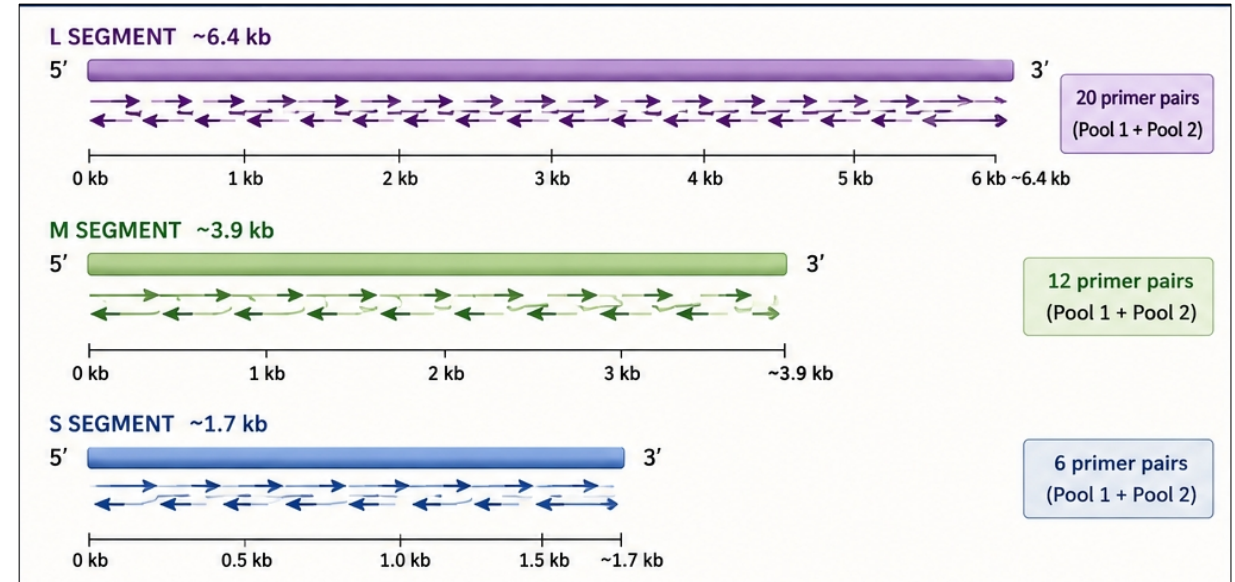
0.5 µg/lane, 8 cm length gel,  
1X TAE, 7 V/cm, 45 min



# Tiled Amplicon Approach: RVFV

# RVFV Multiplex PCR Scheme

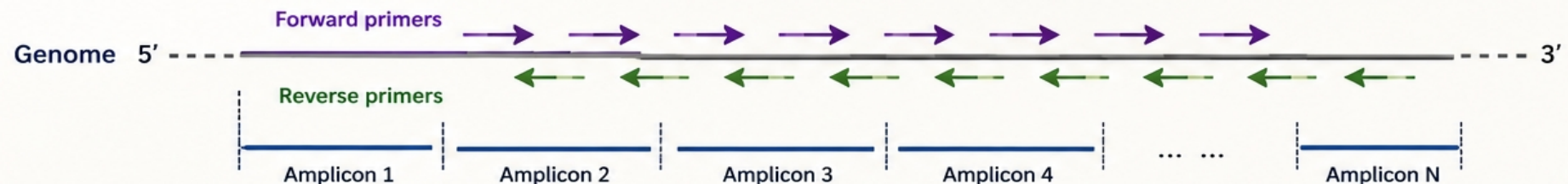
- **74 multiplex primers:** 38 (Pool 1) + 36 (Pool 2)
- Target amplicon size: ~400 bp (Optimal for short-read)
- Coverage by segment:
  - L segment: 20 primer pairs
  - M segment: 12 primer pairs
  - S segment: 6 primer pairs
- In silico predicted coverage: 97-99%



## CONCEPT: Overlapping 'tiled' primers cover entire genome with **NO GAPS**

### TILED PRIMER CONCEPT

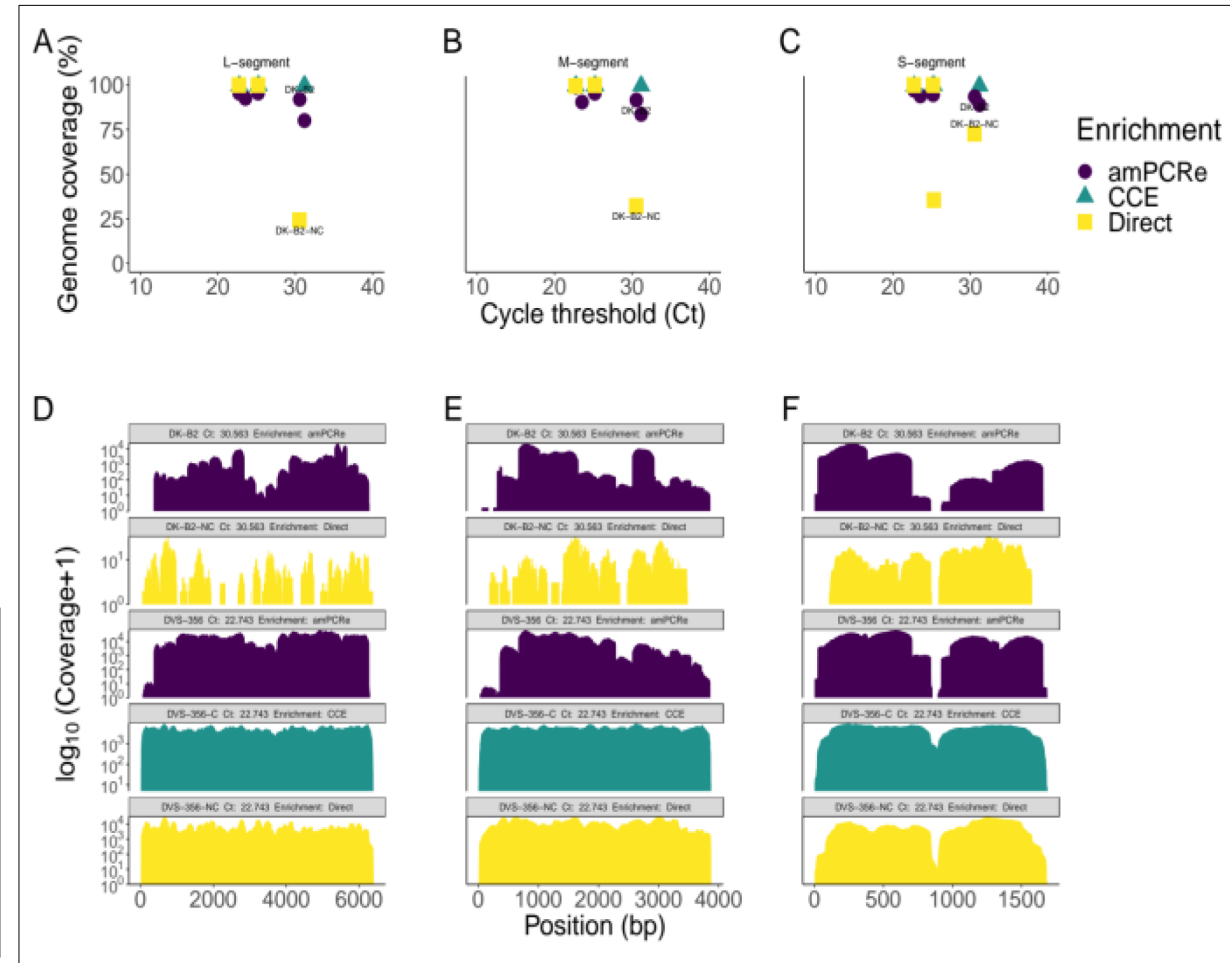
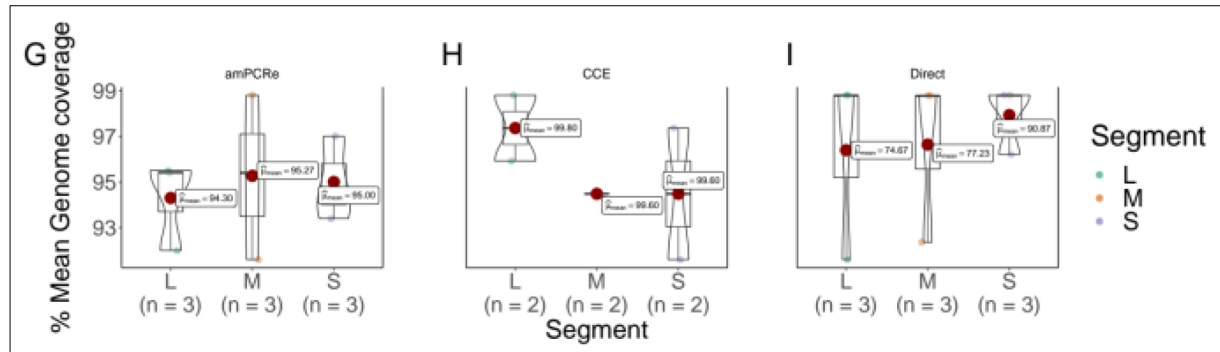
Overlapping amplicons ensure continuous coverage across the entire genome with **NO GAPS**



Amplicons (~400 bp) overlap to ensure continuous coverage with **NO GAPS**

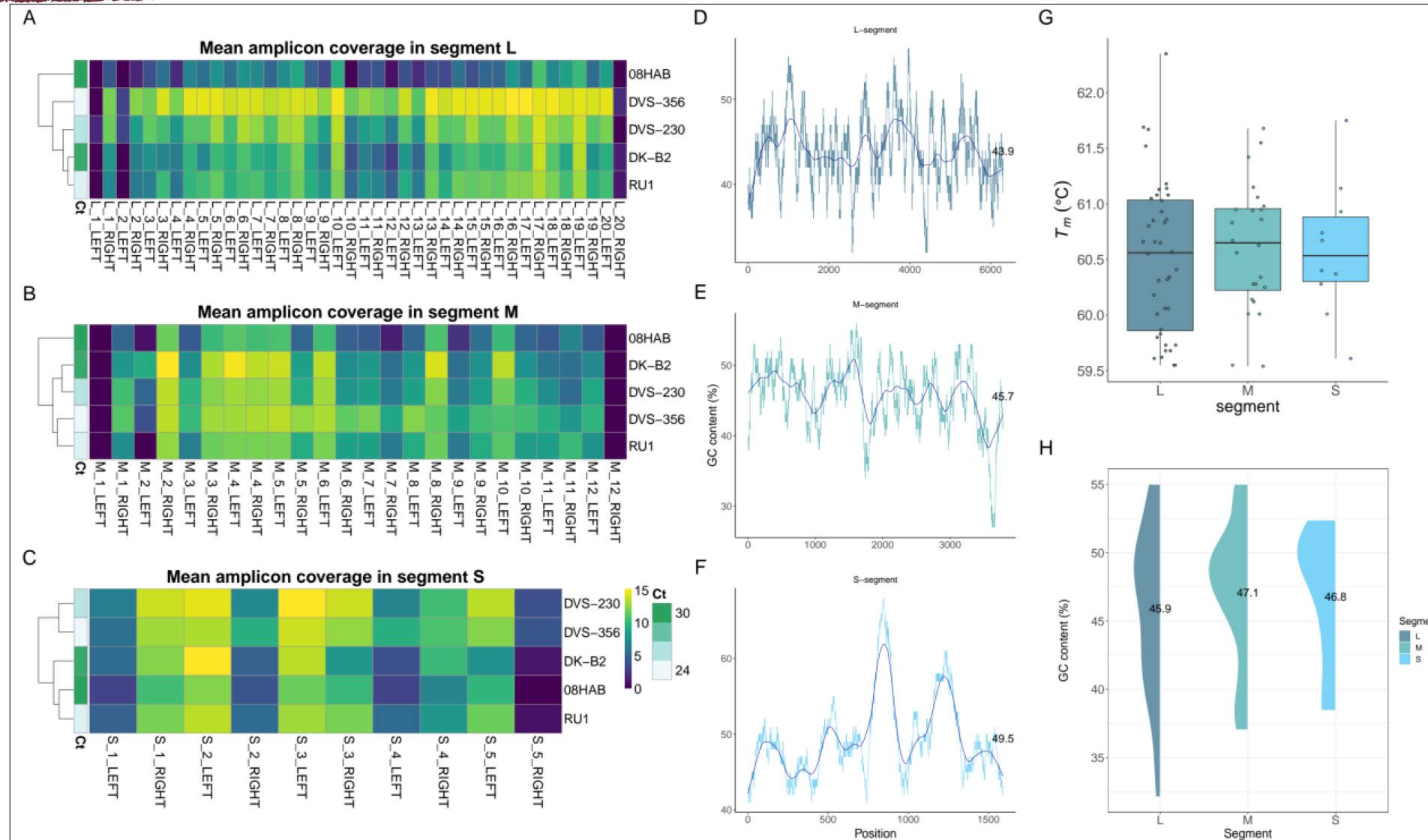
# NGS Approaches: Genome Coverage Comparison

| Study          | Cattle samples - RVF outbreaks (n = 5) |             |                 |
|----------------|----------------------------------------|-------------|-----------------|
| Approaches     | CCE                                    | amPCRe      | Metagenomics    |
| % Coverage     | > 99%                                  | 80 - 97%    | 25 - 70 %       |
| Ct Range       | > 25                                   | > 30        | Failed for > 30 |
| Time           | 7-14 days                              | 24 - 48 hrs | ****            |
| ~ Cost/Isolate | 200 - 500                              | 50 -100     | ****            |



Juma, J.; Konongoi, S.L.; Nsengimana, I.; et al. Using Multiplex Amplicon PCR Technology to Efficiently and Timely Generate Rift Valley Fever Virus Sequence Data for Genomic Surveillance. *Viruses* (2023).

# Tiled Amplicon Approach: Genome Coverage



Juma, J.; Konongoi, S.L.; Nsengimana, I.; et al. Using Multiplex Amplicon PCR Technology to Efficiently and Timely Generate Rift Valley Fever Virus Sequence Data for Genomic Surveillance. *Viruses* (2023).

# Conclusion

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Tiled amplicon sequencing addresses key challenges:

- Enables detection on low viral titer
- Rapid, and
- cost-effective

NGS Whole-Genome Sequencing makes it possible for large-scale surveillance and outbreak response.

# Bioinformatics Analysis: Lineage assignment and phylogenetic analysis

# Why Phylogenetics (In Pathogen Genomics)?

## In outbreak investigation:

- Determine source (local amplification vs. imported)
- Identify transmission chains
- Assess genetic relationship of outbreak isolates
- Detect zoonotic spillover events

## In surveillance:

- Monitor temporal and spatial spread
- Detect introduction of new lineages
- Understand evolutionary dynamics
- Guide control strategy decisions

# Phylogenetic Tree Construction



## Methods used:

- Neighbor-Joining (NJ): Fast, good for large datasets
- Maximum Likelihood (ML): Statistically rigorous, assumes evolutionary model
- Bayesian: Probabilistic, includes uncertainty estimates



## Software tools:

- IQ-TREE: Fast ML trees, bootstrap support
- MrBayes: Bayesian inference, posterior probabilities
- PAUP: Classic phylogenetics software



## Reference sequences:

- 51 unique Gn gene sequences
- 47 unique whole genome sequences
- Chosen for bootstrap support >70%

# Phylogenetic Tree Construction: RVFV

-  Web Application: <https://www.genomedetective.com/app/typingtool/rvf/>
-  Web Application: <https://www.genomedetective.com/db/ui/>
-  Command-line: <https://github.com/ajodeh-juma/rvftyping>

## Raw Data:

1. GenBank SRA - [SRX24403625](https://hpc.ilri.cgiar.org/~gkibet/ndp_vip/) ([https://hpc.ilri.cgiar.org/~gkibet/ndp\\_vip/](https://hpc.ilri.cgiar.org/~gkibet/ndp_vip/))
2. Tutorial: [github - ndp-vip/RVFV Sequence Analysis Workflow](#)
3. [GenBank SRA Search](#) for Other RVFV sequences

## Workflow:

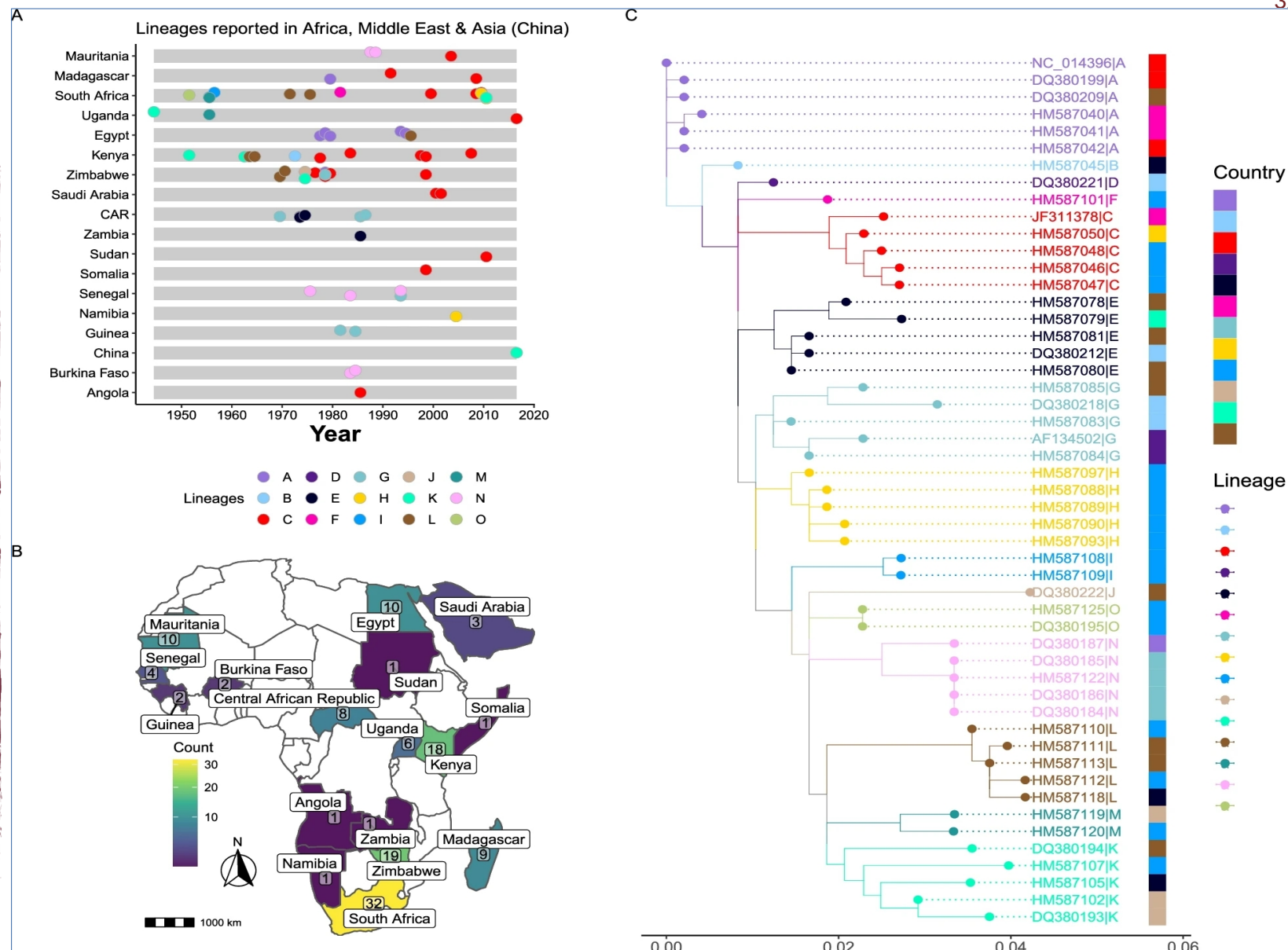
1. User submits sequence (Gn gene or whole genome)
2. DIAMOND search for species classification
3. MAFFT alignment to reference sequences
4. ML tree construction (IQ-TREE)
5. Lineage assignment if bootstrap >70%

## Validation Results (Oyola et al. 2022):

- Gn gene: 100% accuracy on 129 sequences
- Whole genome: 99-100% accuracy on 234 sequences

## What About RVFV?

- Lineages: 15 distinct lineages identified in Africa (A through O)
- Lineage C: Most predominant (documented since 1976)
- Lineage H: Second most common in recent outbreaks
- Other lineages: Geographically or temporally restricted



Juma, J., Fonseca, V., Konongoi, S.L. et al. Genomic surveillance of Rift Valley fever virus: from sequencing to lineage assignment. BMC Genomics 23, 520 (2022).

# Acknowledgements

- Dr Sam Oyola – Senior Scientist
- Dr Bernard Mware
- Dr John Juma - Bioinformatician
- Collins Muli
- Paul Dobi
- Edward Kiritu
- Kennedy Mwangi
- Shebbar Osiany



More about the lecture: <https://github.com/kibet-gilbert/rvf-v-ns-surveillance>

# Tack så mycket!!!

# Asante!!!

## Email & Collaboration:

- Samuel O. Oyola (ILRI): [S.Oyola@cgiar.org](mailto:S.Oyola@cgiar.org)
- John A. Juma (ILRI): [j.juma@cgiar.org](mailto:j.juma@cgiar.org)
- Gilbert Kibet (Presentation Author)

## Online Tools & Databases:

- RVFV Typing Tool: <https://www.genomedetective.com/app/typingtool/rvfv/>
- GitHub/Primal Scheme: [https://github.com/artic-network/primal\\_schemes](https://github.com/artic-network/primal_schemes)
- NCBI GenBank: <https://www.ncbi.nlm.nih.gov/genbank/>

## Key Publications:

- Oyola et al. (2022) BMC Genomics - RVFV lineage assignment tool
- Juma et al. (2023) Viruses - Amplicon PCR methodology

## Training Resources:

- Africa CDC Bioinformatic Training Portal
- Coursera NGS Analysis Courses
- EMBL-EBI Online Workshops